Data Mining Project – Phase 1 Report

1. Domain Understanding

This project focuses on the financial domain, specifically analyzing customer behavior and transaction patterns. Financial institutions rely heavily on data-driven decisions to understand customer spending habits, manage risk, and improve services.

The main objective of this project is to analyze customer data and transaction records to extract meaningful insights about customer behavior, credit utilization, and purchasing patterns. By applying data mining techniques, we aim to identify customer segments, detect unusual activities, and understand relationships between financial attributes.

This analysis is important because it helps businesses:

- Improve customer targeting
- Detect risky or abnormal behavior
- Make better financial decisions

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2. Dataset Description

The project uses two datasets:

Customer Dataset

This dataset contains **387 records** representing customers. It includes features such as:

- Customer ID
- Credit limit (LIMIT)
- Utilized credit (UTILIZED)
- Utilization flags

Transactions Dataset

This dataset contains **9611 records** representing customer transactions. It includes:

- Merchant names
- Product types
- Purchase-related information

The datasets include both **numerical and categorical data**, making them suitable for multiple data mining techniques.

Data Quality Issues

The datasets are not fully cleaned and may contain several issues, including:

- Missing values in some fields
- Duplicate records
- Inconsistent formatting in categorical variables
- Possible outliers in financial values

These issues will be addressed in the data preparation phase.

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3. CRISP-DM Plan

The project will follow the CRISP-DM methodology:

1. Business Understanding

Define the problem of analyzing customer financial behavior to improve decision-making and identify useful insights.

2. Data Understanding

Explore both datasets, analyze distributions, and identify patterns, missing values, and inconsistencies.

3. Data Preparation

Clean the data by handling missing values, removing duplicates, transforming variables, and preparing features for modeling.

4. Modeling

Apply the following techniques:

- Clustering to segment customers
- Classification to predict customer categories
- Regression to analyze relationships between variables
- Anomaly detection to identify unusual transactions

5. Evaluation

Evaluate model performance using appropriate metrics and assess whether results meet the project objectives.

6. Deployment

Develop an interactive dashboard to visualize insights and support decision-making.

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4. Technique Selection Rationale

Different data mining techniques will be used to extract insights from the datasets:

- **Clustering** will be used to group customers based on their financial behavior and spending patterns.
- **Classification** will help predict customer categories or risk levels.
- **Regression** will be applied to study the relationship between credit limits and utilization.
- **Anomaly Detection** will be used to identify unusual or potentially fraudulent transactions.

These techniques are suitable because they address different aspects of the problem and provide a comprehensive analysis of customer behavior.

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5. Environment Setup

The project will be implemented using Python. The following tools and libraries will be used:

- Pandas and NumPy for data processing
- Matplotlib and Seaborn for visualization
- Scikit-learn for machine learning models

A GitHub repository has been initialized to manage the project, including:

- Project structure
- README file
- Version control for collaboration